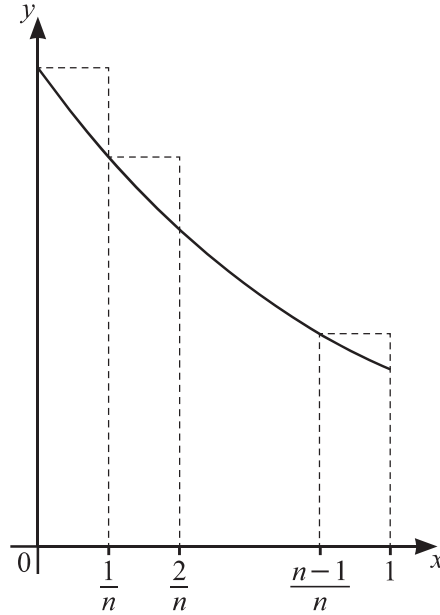




(a) By considering the sum of the areas of these rectangles, show that $\int_0^1 e^{-x} dx < U_n$, where

$$U_n = \frac{e-1}{n(1-e^{-\frac{1}{n}})}. \quad [4]$$

This image shows a full page of white paper with horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the entire width of the page. There are no margins, text, or other markings present.

[illegible][illegible]

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

[illegible]